From Waveguides to Quantum Gates

An Electro-Optical PIC Mach–Zehnder Interferometer

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Purpose: Derive, implement, and validate the link from physical waveguide supermodes to a two-mode quantum optical unitary, matching the operation order of a free-space MZI.

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Abstract

An ideal, lossless Mach–Zehnder interferometer (MZI) can be represented by a product of two-mode unitaries, whether implemented with free-space beamsplitters or waveguide directional couplers. The engineering question is where these matrices come from and how actual device parameters determine their entries. We develop the chain from a scalar electromagnetic supermode eigenvalue problem to a reduced coupled-mode eigenproblem, derive the basis transformation and directional-coupler unitary, and assemble upper-rail input and internal phase shifts with two couplers. The explicitly defined free-space operation sequence is **PRE** → **BS1** → **INTERNAL** → **BS2**; its PIC counterpart is **PRE** → **DC1** → **INTERNAL** → **DC2**. A reproducible worked example shows the Hadamard and identity transformations, and the interactive notebook displays each component matrix, its cumulative product, and single-photon output probabilities. The model is an educational scalar slab-waveguide approximation, not a fabrication-qualified three-dimensional PIC design.

Central result. Once the free-space beamsplitter convention is matched to the directional-coupler port phases, a physical PIC and its free-space counterpart realize the same two-mode optical transformation. The coupling coefficient is derived from electromagnetic supermode splitting rather than chosen as an abstract gate parameter.

Customer discovery

The immediate discovery question is: *where do free-space quantum optical designs encounter uncertainty, delay, or rework when translated into photonic integrated circuits?* The MZI simulator provides a reproducible reference for discussions with free-space optics researchers, PIC designers, quantum hardware teams, and shared quantum testbeds. Discovery should establish which circuit conventions and optical interfaces must be preserved; whether component geometry, wavelength, and voltage controls can be exposed; and what level of electromagnetic fidelity is required before a PIC implementation can be evaluated.

Free-space optics	PIC engineering	Quantum testbeds / users
Target unitaries, port phases, optical layouts	Waveguide supermodes, coupler geometry, EO control	Benchmarks, measurable outputs, verification criteria

Quantum-information target and physical optical layout

A lossless two-port device acts on a single-photon path-encoded state

$$|\psi\rangle = \alpha|1,0\rangle + \beta|0,1\rangle, \qquad \boldsymbol{c} = (\alpha,\beta)^{\mathsf{T}}, \quad \boldsymbol{c}_{\text{out}} = U\boldsymbol{c}_{\text{in}}. \tag{1}$$

Here $|1,0\rangle$ and $|0,1\rangle$ mean one photon in the upper or lower guided spatial mode. Quantum information specifies U; integrated photonic engineering supplies propagation lengths, waveguide separation, refractive indices, and controlled phase shifts that realize it [1, 4, 6]. The device consists of an upper-rail input phase, a first symmetric directional coupler, an upper-rail internal phase, and a second coupler. With column vectors acted upon from the right, the order is

$$\boxed{\text{PRE} \to \text{BS1} \to \text{INTERNAL} \to \text{BS2}, \qquad U_{\text{FS}} = B_2 P_{\text{int}} B_1 P_{\text{pre}}.} \tag{2}$$

The PIC follows the identical physical sequence with B_j replaced by guided-wave couplers C_j. A free-space beamsplitter can use the same complex port-phase convention as C_j; alternative free-space conventions require appropriate input/output port-phase references. Consequently, full complex-matrix comparisons require an explicit common convention; probability agreement alone is weaker.

Physical waveguide model and supermode eigenproblem

Index profile and modal equation

Take two identical parallel slab waveguide cores of width w, edge-to-edge gap g, refractive index n_{core}, embedded in cladding $n_{\text{clad}} < n_{\text{core}}$. For monochromatic TE-like fields in the scalar one-dimensional approximation, let $E_y(x,z) = u(x)e^{-i\beta z}$; the transverse profile solves

$$\left[\frac{d^2}{dx^2} + k_0^2 n^2(x)\right] u(x) = \beta^2 u(x), \qquad k_0 = \frac{2\pi}{\lambda}. \tag{3}$$

The numerical notebook discretizes Eq. (3) on a transverse grid and solves its real symmetric eigenvalue problem for the two highest guided propagation constants. This is a scalar slab model; a full-vector cross-sectional calculation would be needed for a fabrication-level channel-waveguide device [2, 3, 1].

Because $n(x) = n(-x)$ in an ideal symmetric coupler, the eigenfunctions may be selected with definite parity: $u_e(-x) = u_e(x)$ and $u_o(-x) = -u_o(x)$. Their propagation constants, β_e and β_o, are different because their fields overlap differently in the gap. Their splitting produces the optical beating that transfers power between the two rails. The physical geometry and wavelength enter Eq. (3); a coupling angle is *not* supplied independently.

From electromagnetic supermodes to a two-mode matrix

To interpret the supermode calculation in the upper/lower port basis, use the reduced coupled-mode propagation equation

$$i\frac{d}{dz}\begin{pmatrix} a \\ b \end{pmatrix} = \underbrace{\begin{pmatrix} \beta_0 & \kappa \\ \kappa & \beta_0 \end{pmatrix}}_{B} \begin{pmatrix} a \\ b \end{pmatrix}. \tag{4}$$

The common propagation constant β_0 and coupling κ are effective parameters of the two-mode description. Solving $B\boldsymbol{v} = \beta\boldsymbol{v}$ gives

$$\det(B - \beta I) = (\beta_0 - \beta)^2 - \kappa^2 = 0, \quad \beta_e = \beta_0 + \kappa, \quad \beta_o = \beta_0 - \kappa. \tag{5}$$

For β_e, the two equations require $v_1 = v_2$; for β_o, they require $v_1 = -v_2$. Normalizing gives

$$\boldsymbol{v}_e = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{v}_o = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \beta_0 = \frac{\beta_e + \beta_o}{2}, \; \kappa = \frac{\beta_e - \beta_o}{2}. \tag{6}$$

Thus the electromagnetic calculation supplies β_e, β_o and the reduced model reproduces their splitting. The identification of physical input-port amplitudes with orthonormal upper/lower basis states is a two-mode approximation; it becomes less exact when the separated-guide modes overlap strongly.

Deriving the waveguide-coupler unitary

Basis transformation: no extra hardware gate

The columns of the matrix S are the normalized supermode eigenvectors, expressed in the upper/lower port basis:

$$S = (\boldsymbol{v}_e\ \boldsymbol{v}_o) = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \quad \begin{pmatrix}a\\ b\end{pmatrix} = S\begin{pmatrix}c_e\\ c_o\end{pmatrix}, \quad \begin{pmatrix}c_e\\ c_o\end{pmatrix} = S^\dagger\begin{pmatrix}a\\ b\end{pmatrix}. \tag{7}$$

$S^\dagger$ appears because orthonormal eigenvectors make S unitary: $S^{-1} = S^\dagger$. This is a mathematical coordinate change, *not* a physically inserted Hadamard gate. A photon injected into the upper guide corresponds to $(1,0)^\mathsf{T}$ in port coordinates and $(1,1)^\mathsf{T}/\sqrt{2}$ in supermode coordinates.

Propagation in the eigenmode basis

The eigenmodes propagate independently, so over the coupled length L their coefficients acquire phases

$$D(L) = \begin{pmatrix}e^{-i\beta_e L} & 0\\ 0 & e^{-i\beta_o L}\end{pmatrix}, \qquad \boldsymbol{c}_{\rm port}(L) = \underbrace{SD(L)S^\dagger}_{U_{\rm DC}(L)}\boldsymbol{c}_{\rm port}(0). \tag{8}$$

Writing $\bar\beta = (\beta_e + \beta_o)/2$ and $\theta = \kappa L$ and carrying out the matrix product yields

$$\begin{aligned} U_{\rm DC}(L) &= \frac{1}{2}\begin{pmatrix}e^{-i\beta_e L} + e^{-i\beta_o L} & e^{-i\beta_e L} - e^{-i\beta_o L}\\ e^{-i\beta_e L} - e^{-i\beta_o L} & e^{-i\beta_e L} + e^{-i\beta_o L}\end{pmatrix} \\ &= e^{-i\bar\beta L}\underbrace{\begin{pmatrix}\cos\theta & -i\sin\theta\\ -i\sin\theta & \cos\theta\end{pmatrix}}_{C(L)}. \end{aligned} \tag{9}$$

The simulator drops the common $e^{-i\bar\beta L}$ phase for each ideal coupler; differential phases between interferometer rails are accounted for explicitly. The convention is $e^{-i\beta z}$, consistent with the $-i$ cross-coupling amplitudes. For input $|1,0\rangle$,

$$|1,0\rangle \longmapsto \cos(\kappa L)|1,0\rangle - i\sin(\kappa L)|0,1\rangle, \quad P_{10} = \cos^2(\kappa L),\ P_{01} = \sin^2(\kappa L). \tag{10}$$

At $\kappa L = \pi/4$ both outputs have probability $1/2$. No photon is divided into fractional photons: Eq. (10) is a superposition of alternatives for detecting the whole photon.

Electro-optical phases and composite MZI

The upper-rail input pre-phase and upper-rail internal phase are, respectively,

$$P_{\rm pre}(\chi) = \begin{pmatrix}e^{i\chi} & 0\\ 0 & 1\end{pmatrix}, \quad P_{\rm int}(\phi) = \begin{pmatrix}e^{i\phi} & 0\\ 0 & 1\end{pmatrix}, \quad \phi = \phi_{\rm bias} + \pi\frac{V}{V_\pi}. \tag{11}$$

The V_π parameter is the half-wave voltage for the chosen effective differential-phase response; this calibrated relation does not itself solve the electro-optic material problem. The device order, now written in physically meaningful PIC components, is

$$\boxed{\text{PRE} \to \text{DC1} \to \text{INTERNAL} \to \text{DC2}, \qquad U_{\rm PIC} = C_2 P_{\rm int}(\phi) C_1 P_{\rm pre}(\chi).} \tag{12}$$

Each C_j is calculated from its own gap, length and supermode eigenvalue splitting. Free-space beamsplitters B_1, B_2 with $B_j = C_j$ in the adopted port-phase convention obey $U_{\rm FS} = U_{\rm PIC}$ by Eqs. (2) and (12), with no requirement that their hardware geometry be identical. The same 2×2 unitary therefore describes both implementations in the specified single-frequency, two-mode, ideal-lossless model.

Worked example: Hadamard and identity benchmarks

Geometry → eigenvalues → balanced coupler

For both couplers choose $w = 0.75\,\mu$m, $g = 0.35\,\mu$m, $n_{\mathrm{core}} = 1.50$, $n_{\mathrm{clad}} = 1.45$, and $\lambda = 1.55\,\mu$m. The notebook's scalar eigenvalue solver produces the following values (full internal precision is retained):

Parameter	Computed value
Even supermode β_e	$5.968509\,\mu\mathrm{m}^{-1}$
Odd supermode β_o	$5.865574\,\mu\mathrm{m}^{-1}$
Supermode splitting 2κ	$0.102935316\,\mu\mathrm{m}^{-1}$
Coupling coefficient κ	$0.051467658\,\mu\mathrm{m}^{-1}$
Exact 50:50 length $L_{50:50} = \pi/(4\kappa)$	$15.260033\,\mu$m

Using Eqs. (6)–(9), both balanced directional couplers become

$$C_1 = C_2 = C = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}. \tag{13}$$

Hadamard target

Choose $\chi = 0$, $V = -2.50$ V, $V_\pi = 5.00$ V and $\phi_{\mathrm{bias}} = 0$, so $\phi = -\pi/2$. By Eqs. (11), (12), and (13),

$$\begin{aligned} U_H &= CP_{\mathrm{int}}(-\pi/2)CP_{\mathrm{pre}}(0) \\ &= \begin{pmatrix} -(1+i)/2 & -(1+i)/2 \\ -(1+i)/2 & (1+i)/2 \end{pmatrix} = e^{-3\pi i/4}\underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{H}. \end{aligned} \tag{14}$$

Thus the complex matrix, not only its output intensities, equals the free-space Hadamard target up to an unobservable global phase. An input $|1,0\rangle$ yields $P_{10} = P_{01} = 1/2$.

Identity target

Keeping the *same two balanced couplers*, set $\phi = \pi$ ($V = +5.00$ V, with $V_\pi = 5.00$ V and zero bias) and $\chi = \pi$. Then

$$CP_{\mathrm{int}}(\pi)C = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad U_I = CP_{\mathrm{int}}(\pi)CP_{\mathrm{pre}}(\pi) = I_2. \tag{15}$$

Both basis inputs now emerge from their corresponding output ports with unit probability. If χ is left at zero, the same internal setting gives a diagonal phase gate rather than identity. The two phase locations cannot be silently interchanged.

Simulator, reproducibility, and numerical acceptance

The companion `sim_PIC_MZI_v2.ipynb` uses physical controls for the slab profile and coupler lengths, plus χ, V, V_π, and bias. For each stage (input, PRE, DC1, INTERNAL, DC2), it reports the element matrix, the *left-multiplied cumulative product*, complex single-photon amplitudes, and probabilities. A separate display highlights U_{PIC} and the phase-aligned Hadamard matrix.

For unitary matrices U and target T, the global-phase-insensitive comparison used for the Hadamard benchmark is

$$\varepsilon_T = \min_{\gamma} \|e^{-i\gamma}U - T\|_F, \qquad \gamma = \arg\mathrm{Tr}(T^\dagger U), \qquad \varepsilon_{\mathrm{unit}} = \|U^\dagger U - I\|_F. \tag{16}$$

A rounded manual length of $15.2600\,\mu$m generates a small but finite Hadamard residual (observed 3.42×10^{-6}), whereas the precise length computed by the **Set exact Hadamard reference** action gives a machine-precision comparison in the regression tests. This is a settings-precision difference, not nonunitarity. The user-reported run with rounded length had composite unitarity residual 2.345×10^{-16}.

The included benchmark cell checks component and composite unitarity, stage ordering, independently evaluated analytical products, pre-phase behavior, interference, waveguide-gap response, and the Hadamard and identity targets. The result is an idealized test of the mathematical and software implementation; device fabrication tolerances, vectorial electromagnetic effects, scattering losses, and photon distinguishability are outside this simulator's scope.

Outlook

An optical circuit specified by its target scattering matrix can be translated into an integrated design by selecting directional-coupler lengths, waveguide dimensions, and phase settings that implement the same matrix under a consistent port convention. The present two-mode MZI is a physically parameterized and independently validated building block. Extension to meshes of MZIs would connect free-space interferometer design to scalable PIC architectures, as developed in the optical unitary-decomposition literature [5, 7]; engineering-ready translation will require device-specific electromagnetic modeling and experimental calibration.

Reading resources and foundational papers

References

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems* (Cambridge University Press, 2015). doi:10.1017/CBO9781316084168.

[2] G. T. Reed and A. P. Knights, *Silicon Photonics: An Introduction* (Wiley, 2004).

[3] M. A. G. Abushagur, *Applied Photonics: An Introduction for Physicists and Engineers* (Springer, 2025).

[4] K. Thyagarajan, *Integrated Quantum Photonics* (Springer, 2025).

[5] M. Reck, A. Zeilinger, H. J. Bernstein, and P. Bertani, "Experimental realization of any discrete unitary operator," *Phys. Rev. Lett.* **73**, 58–61 (1994). doi:10.1103/PhysRevLett.73.58.

[6] A. Politi, M. J. Cryan, J. G. Rarity, S. Yu, and J. L. O'Brien, "Silica-on-silicon waveguide quantum circuits," *Science* **320**, 646–649 (2008). doi:10.1126/science.1155441.

[7] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, "Optimal design for universal multiport interferometers," *Optica* **3**, 1460–1465 (2016). doi:10.1364/OPTICA.3.001460.